

Providing Diverse Opportunities for Capstone Projects in Biomedical Engineering

Mansoor Nasir, Eric G. Meyer, Yawen Li

Biomedical Engineering, Lawrence Technological University



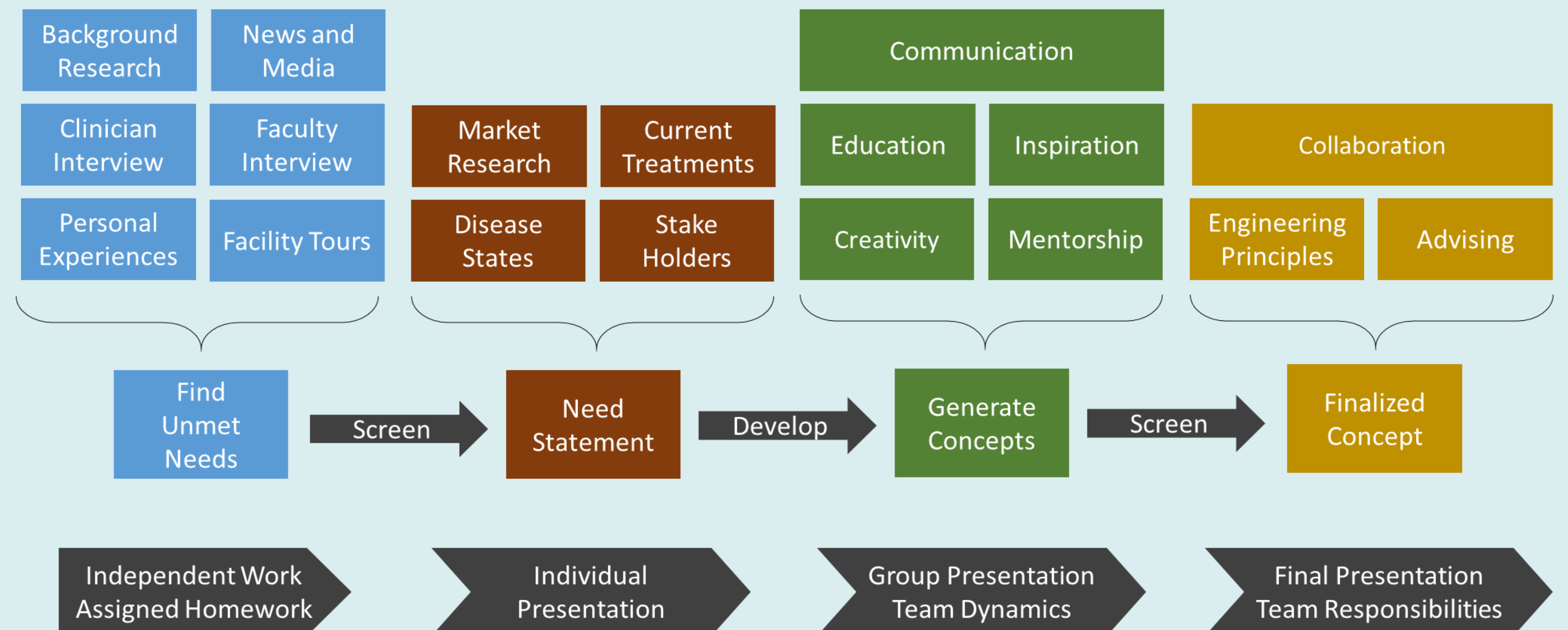
Capstone Senior Project Sequence

Course	Semester	Cr.	Focus
BME 4013	Fall	3	Ideation
BME 4022	Spring	2	Implementation

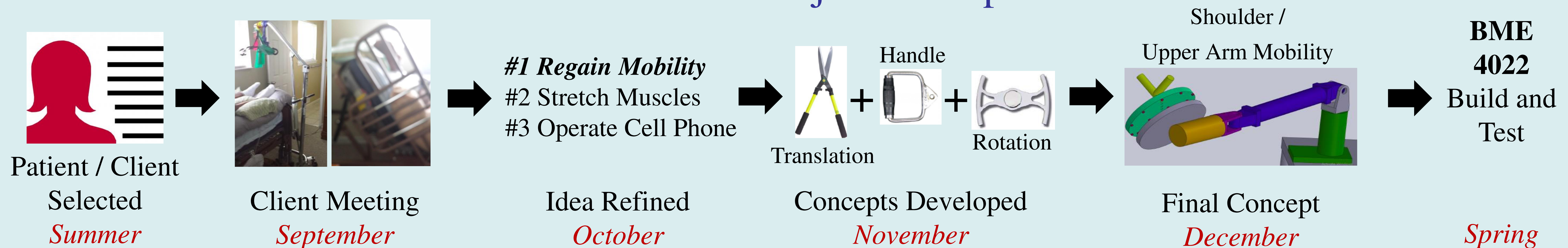
Preparatory Coursework

Course	Semester	Cr.	Focus
BME 3002	Spring (Soph.) Fall (Junior)	2	Best Practices Regulation, Ethics
BME 4113	Spring (Junior)	3	Med. Device Design Ideation Process

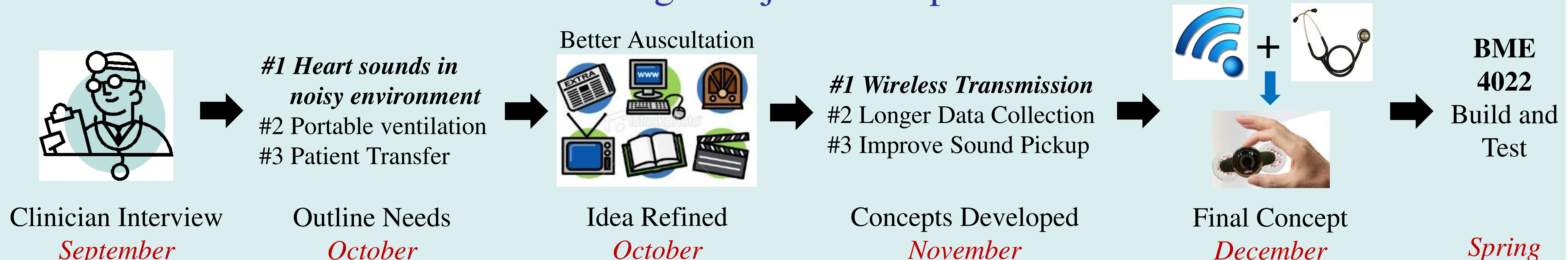
Ideation Process (BME 4013)



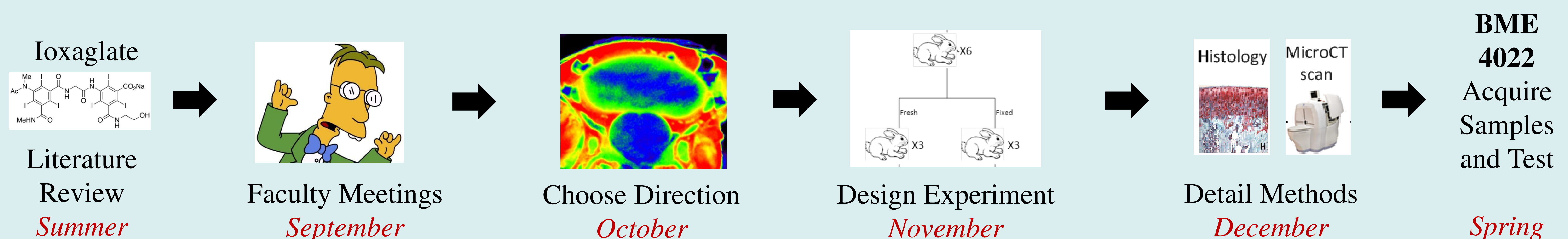
Patient / Client Project Example



Design Project Example



Applied Research Project Example



Project I – Faculty Preparations

- Schedule Speakers:
 - Industrial Sponsors
 - Clinicians / Hospital Visits
 - BME and Engineering Faculty Speakers
 - Graduating Seniors to Speakers
 - Electronic Resources
- Material Resources / Equipment
- Grading Rubrics
- Examples of Previous Projects
- Mentors / Advisors

Helpful Assignments / Activities

- Talk to a student in different department (Homework)
- Interview a clinician (Homework)
- Patent and 510k search (Homework)
- Same Problem – Different Solutions (In Class)
- Develop a Need Statement (In Class)
- Role Playing (In Class)

Project II – Faculty Preparation

- Equipment Availability
- Team Dynamics and Responsibilities
- Mentor / Advisor Schedules
- Getting Student Buy-in
- Intermediate Goals and Task Lists